

Follow-up of Endodontically Treated Teeth without Crown Coverage in Vidarbha Region: A Long-Term Retrospective Cohort Study

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INTRODUCTION

An occlusal coverage with endodontically treated teeth (ETTs) is considered an optimum treatment. There are multiple reasons of loss of root canal (RC)-treated teeth as reported by Caplan and Weintraub. Two factors are considered for the failure of

RC-treated teeth as nonvital condition of teeth and loss of dentin as tooth structure.

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Papa *et al.* reported that there is no significant difference in moisture content of vital and nonvital teeth. Their matched pair study carried out by Sedgley and Messer concluded that teeth post endodontic treatment do not become brittle. The necessity to place a crown on an ETT has been investigated in clinical trials with limited observation time, but a Cochrane review does not reveal any conclusion due to lack of suitable studies.

In the 3-year follow-up of ETTs by Nikhade *et al.*, it was observed that coronal coverage of ETTs presumably reduced the risk of fracture or other failures. The result showed the favorable survival of ETTs and restoration level. The crown restoration has been a suggested protocol to reinforce or strengthen a tooth after endodontic treatment.

The use of various direct restorative materials and techniques in compromised teeth has shown a good prognosis *in vitro*.

Table 1: Characteristics of 148 teeth in 300 patients

Variable	All patients (n=300), n (%)
Gender	
Men	93 (62.83)
Women	55 (37.16)
Age (years)	19-67
Preoperative factors, tooth-based all teeth (n=148)	
Jaw	
Lower	53 (35.81)
Upper	95 (64.18)
Teeth	
Incisor	13 (8.78)
Canine	7 (4.72)
Premolar	52 (35.13)
Molar	76 (51.35)
Presence of opposing teeth	
Natural tooth	80 (54.05)
Crown	48 (32.43)
Extracted	20 (13.51)

Table 2: Number of events

Type of filling	Total (n)	Number of events	Censored, n (%)
Composite	99	99	0
Silver amalgam	32	32	0
GIC	17	17	0
Overall	148	148	0

GIC: Glass-ionomer cement

Reeh *et al.* studied the prospective results for different restorative techniques in nonvital teeth. Standardized MOD cavities were used. The study analyzed that composite resins with etching showed a relative stiffness near to that of the unaltered tooth. Silver amalgam restoration showed little or no resistance to cuspal movement, whereas a gold onlay extended greater resistance to movement than that of the unaltered tooth.

In a 20-year retrospective study, the only prospective study that compared performance between the uncrowned and crowned ETTs was performed by Mannocci *et al.*

The aims of this study were to evaluate the criteria for survival of ETTs without crown coverage and to identify possible explanations for the differences observed.

For RC-treated teeth, having a large restoration which is not easily replaceable, the survival of restoration and tooth is questionable as in such cases the complications such as failure of RC treatment as well as vertical fractures are common resulting in tooth loss.

The general condition of ETTs is such that they are restored with complex as well as extensive restorations where repair or replacement is difficult to perform, thus long-term follow-ups are mandatory. When retrospective data from the patient files can be retrieved, an analytical study of factors contributing to longevity is possible. Many clinical studies pertaining to ETTs are done to assess the longevity.

The present observational retrospective study aimed to justify the long-term longevity of ETTs without crown coverage in the patients in Vidarbha region of Maharashtra, India. The parameters were added to the previous study carried out and assessed after 3 years. The same patients were evaluated for 10 years to determine the survival of ETTs.

Cohort

This retrospective study used the records from the routine patients coming to the outpatient department in the dental hospital located in Vidarbha region of Maharashtra, India. This region consists of 26 districts connected with each other. Data were collected from 2010 to 2020. Follow-up of the first 3 years was done till 2013, and the analysis was done for the survival and success of ETTs using Kaplan–Meier statistics and was published.

MATERIALS AND METHODS

The status of the patients was updated in terms of patient intent to continue with further observation every year. The

status of restoration was updated with consent of patients, and the present condition of the tooth in question was observed. Potential carious lesions, questionable crown or root fractures, loss of restoration, and tooth loss were taken into consideration. All the previous records were reviewed for basic data collected and viewed after 3 years. Recall program was scheduled for every 6 months. Records from patients who regularly visited the dental hospital for the presence of ETTs were as follows:

1. A restoration was placed at least 6 months before the last recall visit [Graphs 1-2]
2. A restoration was placed within 6 months after endodontic treatment
3. Records contained information on the ETTs and the dentition.

Screening of the patient files yielded 450 ETTs. From these, 300 RC-treated teeth were evaluated which met the inclusion criteria. The following data were collected from the patient's records.

A checklist chart was made according to the guidelines given in the study carried out by Skupien JA *et al.* to record the characteristics of patient's ETT.

1. Characteristics of the involved tooth and the dentition including the number of teeth, its type, presence of caries while entering the hospital, and decayed/missing/filled teeth at the time of endodontic treatment

2. The data of endodontic treatment which includes relevant information such as number of canals, number of sessions, and number of filled canals
3. The date of placement of the follow-up restoration, the type of restoration, number of surfaces, placement of a post, and core buildup
4. The date and the type of all interventions on the ETT in the period after endodontic treatment.

Dental records were assessed for baseline data on the date of endodontic completion, previous changes in tooth condition. The patients were on 6 months, and 1, 2, and 5 years till 10.3 years post endodontic recall program. Though not all continued to attend. Patients were also enquired about any changes in tooth condition. Only permanent teeth which had not received provisional crowns, definitive restorations with cuspal coverage, post and core restorations, and/or crowns at the time of recall were included. Treatment outcome was defined as a failure if there were negative findings in the condition of a tooth that required a restoration, tooth repair, or extraction. Failure characteristics of tooth changes included recurrent caries at the margin of a restoration, crack/fracture line, loss of restoration, fracture of tooth or restoration, and/or vertical root fracture. ETTs that were lost because of endodontic and periodontal reasons were also excluded. The independent variables assessed were age of patient, gender, location of tooth in arch, existence of opposing dentition, adjacent teeth, remaining tooth structure, and type of restorative material.

Three hundred RC-treated teeth in 148 subjects that satisfied the inclusion criteria were included in this study. The age range of the subjects was between 20 and 45 years.

The observation period ranged from 6 months to 10.3 years. At the end of 10 years, 93 teeth were found to be intact with restorations without full coverage. The follow-up began

from the date of endodontic therapy completion and ended on the date of negative findings in a tooth condition and/or tooth extraction. For teeth that were not determined a failure, follow-up ended on the date of data collection in the clinic [Tables 1-4].

CONCLUSION

The limitation of this study is the nonrandom allocation of treatments to the teeth. The treatment was decided by the clinician and patient based on the specific clinical case. Another limitation was the fact that the number of teeth in each group of the study variables was not well distributed and the observation periods were not equal. In this study, data were not collected on the quality of endodontic treatment itself based on radiographic information, and it will be a subject of a further study. The advantage of the present study is that the status and survival of the ETTs with various restorations reflect the situation in routine practice and provide information about the future of certain restorative materials. The result of the present study shows that restoration on ETTs showed favorable survival on tooth and restoration level.

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Conflicts of interest

There are no conflicts of interest.